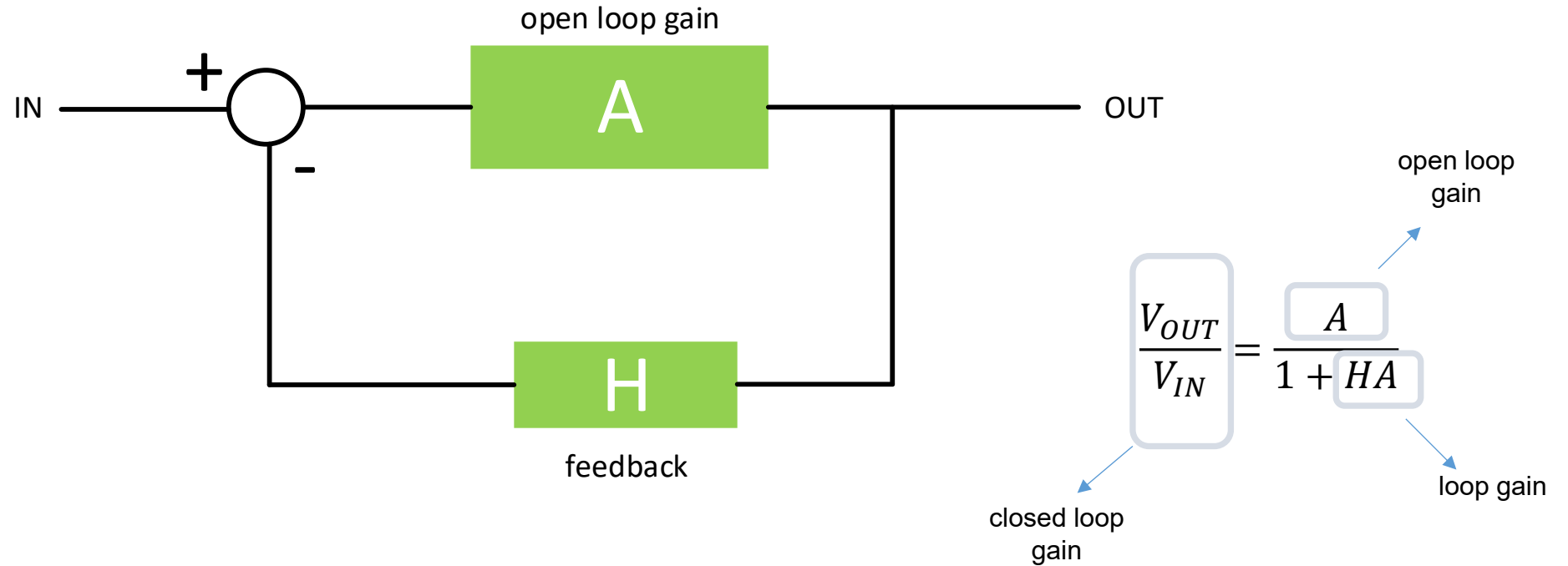


Introduction to Feedback Theory and Frequency Response

BASIC FEEDBACK THEORY



$$V_{OUT} = A (V_{IN} - HV_{OUT})$$

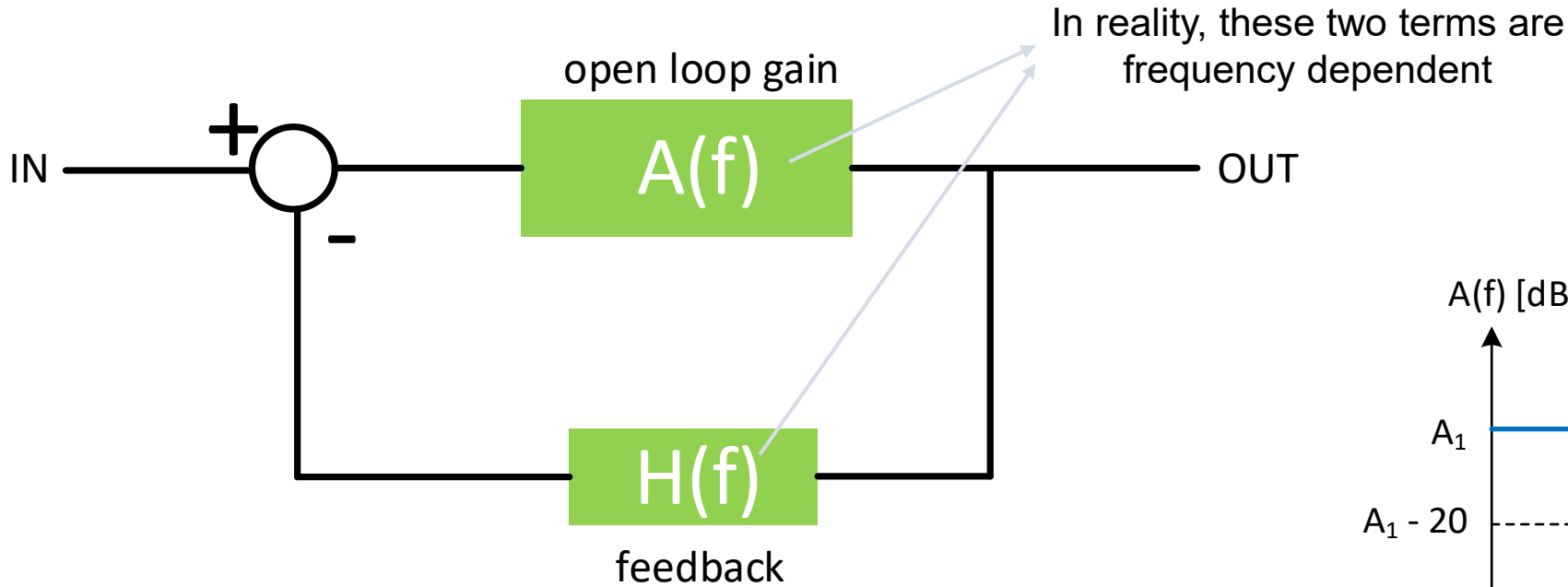
$$\frac{V_{OUT}}{V_{IN}} = \frac{A}{1 + HA} \quad \text{If } A \rightarrow \infty$$

$$\frac{V_{OUT}}{V_{IN}} = \frac{1}{H}$$

closed loop gain

In reality, open loop gain is not stable due to temperature, non-linearity etc. With negative feedback, the closed loop gain can be stabilized since the gain depends only on the negative feedback factor

BASIC FEEDBACK THEORY



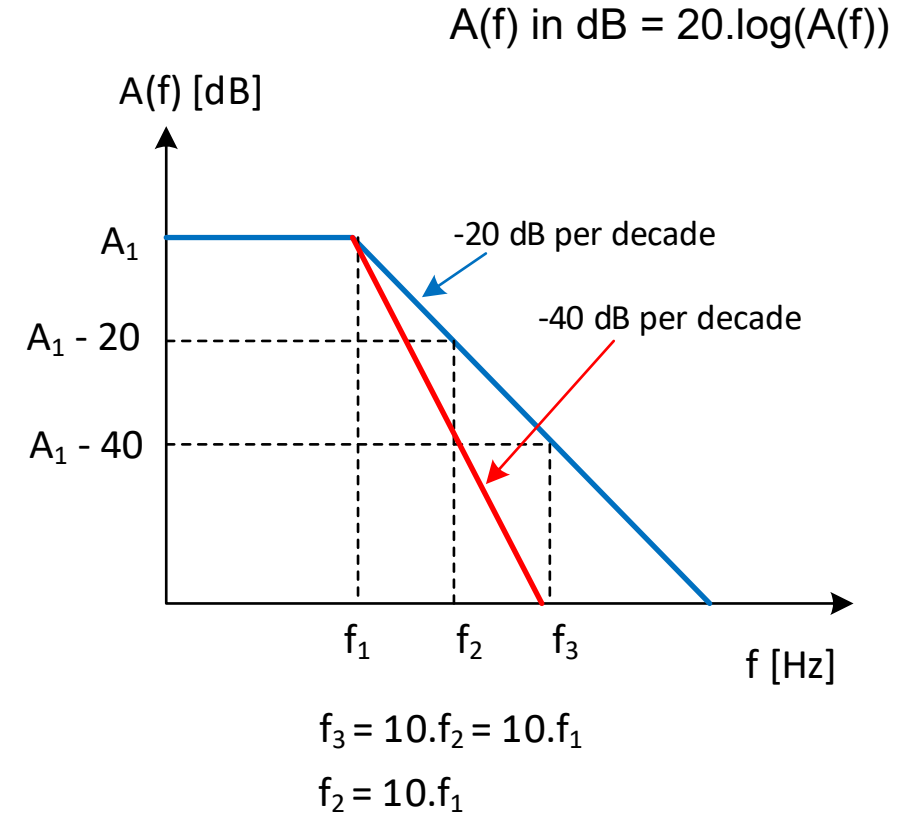
$$V_{OUT}(f) = A(f) (V_{IN}(f) - H(f)V_{OUT}(f))$$

closed loop gain

$$\frac{V_{OUT}(f)}{V_{IN}(f)} = \frac{A(f)}{1 + H(f)A(f)}$$

open loop gain

loop gain



Please note that $V_{OUT}(f)/V_{IN}(f) \rightarrow \infty$, if $H(f)A(f) = -1$ at a certain frequency

In other words, $V_{OUT}(f)/V_{IN}(f)$ **becomes unstable** $\longrightarrow$ Negative feedback must be implemented carefully!

BASIC FEEDBACK THEORY

closed loop gain $\leftarrow \frac{V_{OUT}(f)}{V_{IN}(f)} = \frac{A(f)}{1 + H(f)A(f)} \rightarrow$ open loop gain

loop gain

frequency domain

closed loop gain $\leftarrow \frac{V_{OUT}(s)}{V_{IN}(s)} = \frac{A(s)}{1 + H(s)A(s)} \rightarrow$ open loop gain

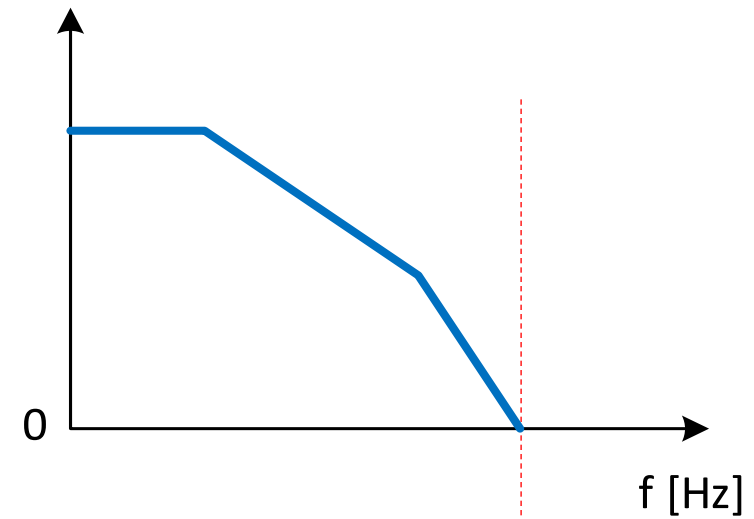
loop gain

s domain

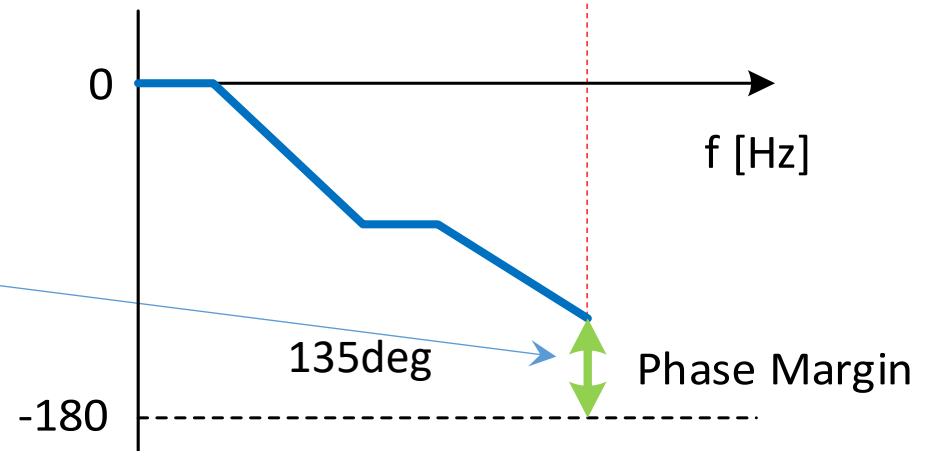
Please note that $V_{OUT}(f)/V_{IN}(f) \rightarrow \infty$, if $H(f)A(f) = -1$ or in other words, loop gain at a certain frequency

A larger phase margin implies a more stable system

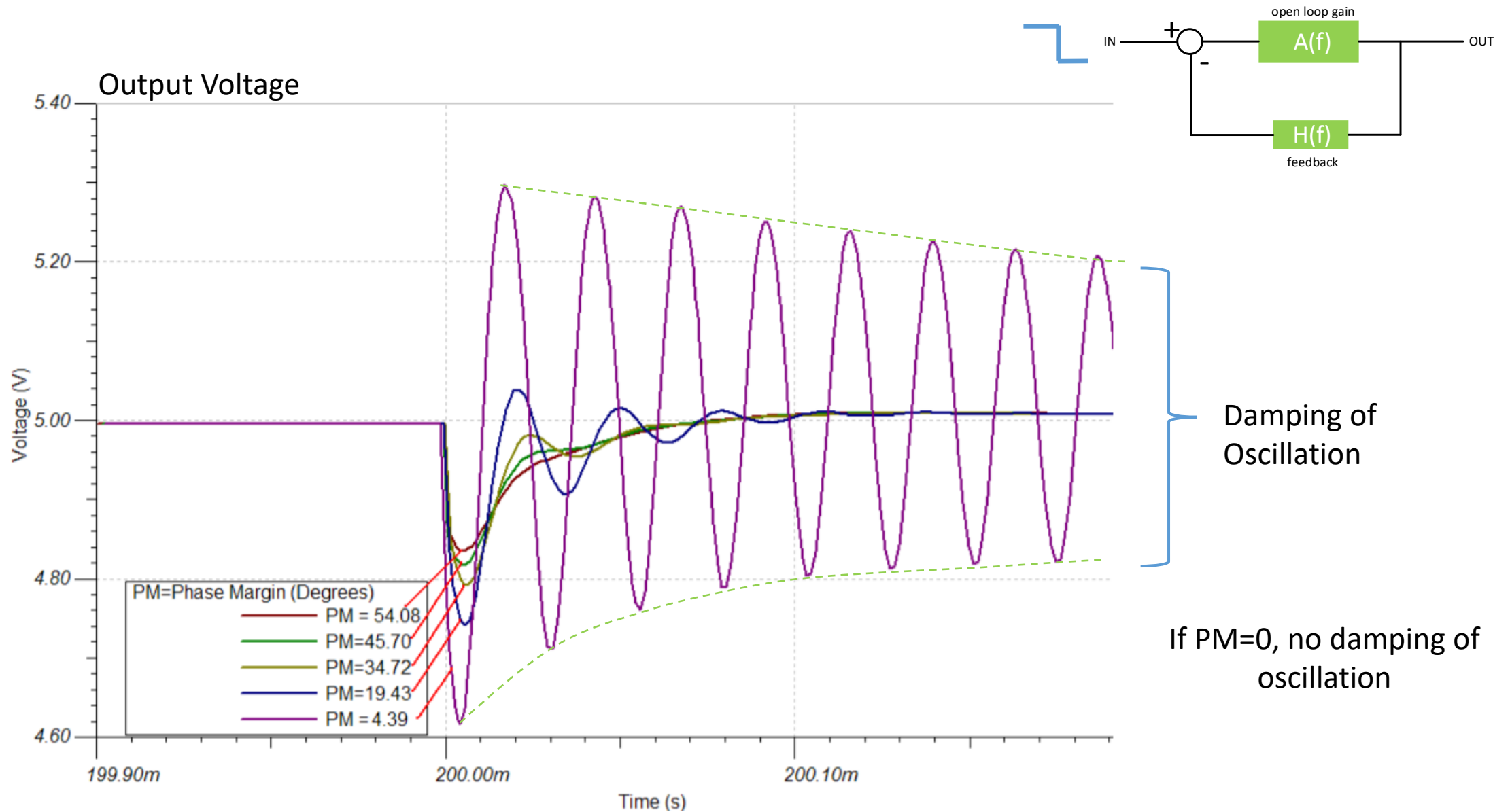
$|A(f)H(f)|$ [dB]



$A(f)H(f)$ [°]



BASIC FEEDBACK THEORY



References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.